INERTIA

PHY 101

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WHAT IS A GYROSCOPE

Gyroscope

- This a device that measures and maintains orientation and angular velocity often using a spinning wheel or a beam of light that resists changes to its axis of rotation
- Figure 1, shows a gyroscope, defined as a spinning disk in which the axis of rotation is free to assume any orientation. When spinning, the orientation of the spin axis is unaffected by the orientation of the body that encloses it. The body or vehicle enclosing the gyroscope can be moved from place to place and the orientation of the spin axis will remain the same.
- Gyroscopes are very useful in navigation, especially where magnetic compasses can't be used, such as in manned and unmanned spacecraft, intercontinental ballistic missiles, unmanned aerial vehicles, and satellites like the Hubble Space Telescope.



Figure 1: A gyroscope consists of a spinning disk about an axis that is free to assume any orientation.

- We illustrate the **precession** of a gyroscope with an example of a top in the next two figures. If the top is placed on a flat surface near the surface of Earth at an angle to the vertical and is not spinning, it will fall over, due to the force of gravity producing a torque acting on its center of mass. This is shown in Figure 2.
- However, if the top is spinning on its axis, rather than topple over due to this torque, it precesses about the vertical, shown in 3. This is due to the torque on the center of mass, which provides the change in angular momentum.

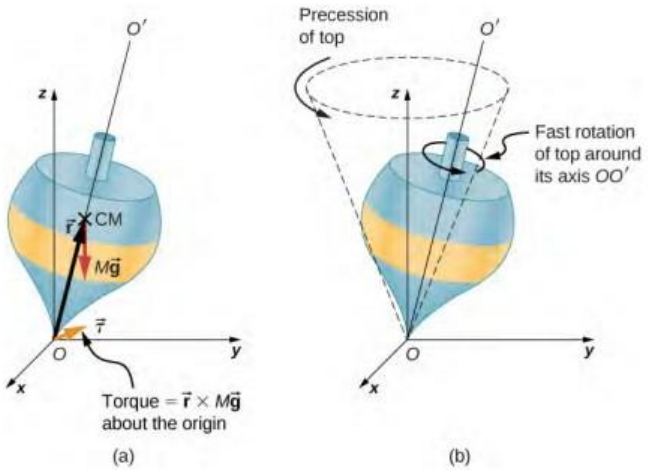
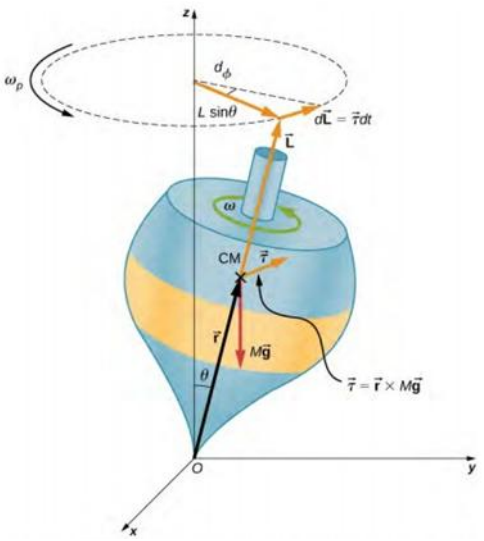


Figure 2: If the top is not spinning, there is a torque $\vec{r} \times M\vec{g}$ about the origin, and the top falls over. (b) If the top is spinning about its axis OO′, it doesn't fall over but precesses about the z-axis.

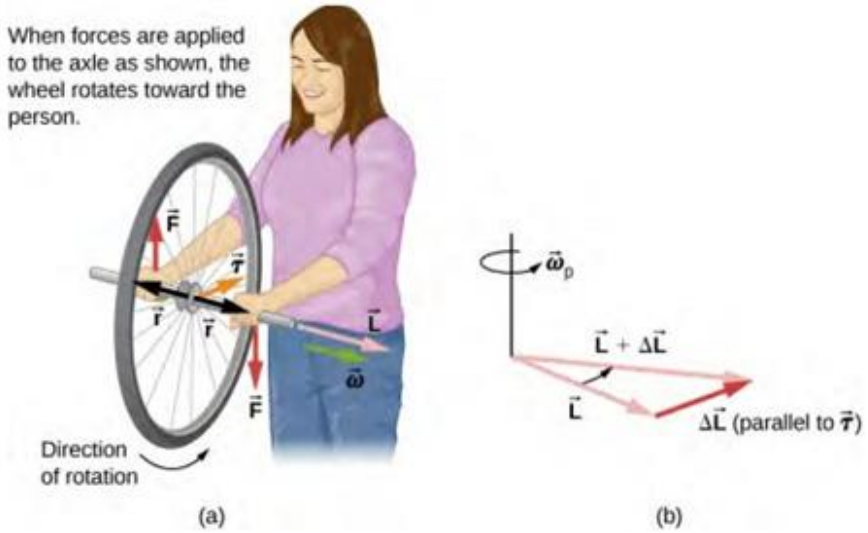
Figure 3 shows the forces acting on a spinning top. The torque produced is perpendicular to the angular momentum vector. This changes the direction of the angular momentum vector $\vec{L}$ according to $d\vec{L} = \vec{\tau}\, dt$, but not its magnitude. The top precesses around a vertical axis, since the torque is always horizontal and perpendicular to $\vec{L}$. If the top is **not** spinning, it acquires angular momentum in the direction of the torque, and it rotates around a horizontal axis, falling over just as we would expect.

- Figure 3: The force of gravity acting on the centre of mass produces a torque $\vec{\tau}$ in the direction perpendic to $\vec{L}$. The magnitude of $\vec{L}$ doesn't change but its direction does, and the top precesses about the z-axi



- We can experience this phenomenon first hand by holding a spinning bicycle wheel and trying to rotate about an axis perpendicular to the spin axis. As show Figure 4, the person applies forces perpendicular to the spin axis in an attempt to rotate the wheel, but instead, the wheel axis starts to change direction to her left due to the applied torque.

- Figure 4: (a) A person holding the spinning bike wheel lifts it with her right hand and pushes down with her left hand in an attempt to rotate the wheel. This action creates a torque directly toward her. This torque causes a change in angular momentum $\Delta\vec{L}$ in exactly the same direction. (b) A vector diagram depicting how $\Delta\vec{L}$ and $\vec{L}$ add, producing a new angular momentum pointing more toward the person. The wheel moves toward the person, forces she exerts on it.



- We all know how easy it is for a bicycle to tip over when sitting on it at rest. But when riding the bicycle at a good pace, it is harder to tip it over because we must change the angular momentum vector of the spinning wheels.
- Also, when a spinning disk is put in a box such as a Blu-Ray player, try to move it. It is easy to translate the box in a given direction but difficult to rotate it about an axis perpendicular to the axis of the spinning disk, since we are putting a torque on the box that will cause the angular momentum vector of the spinning disk to precess.

We can calculate the precession rate of the top in Figure 3 and we see that the magnitude of the torque is

$$\tau = rMg\sin\theta. \qquad (1)$$

$$dL = rMg\sin\theta dt. \qquad (2)$$

$$\frac{dL}{L\sin\theta} \quad \frac{rMg\sin\theta}{L\sin\theta} \quad \frac{rMg}{L}$$

ϕ

ϕ

ω

$\omega,$

r

M

g

- In this derivation, we assumed that $\omega_P << \omega$, that is, that the precession angular velocity is much less than the angular velocity of the gyroscope disk. The precession angular velocity adds a small component to the angular momentum along the z-axis. This is seen in a slight bob up and down as the gyroscope precesses, referred to as nutation.
- Earth itself acts like a gigantic gyroscope. Its angular momentum is along its axis and currently points at Polaris, the North Star. But Earth is slowly precessing (once in about 26,000 years) due to the torque of the Sun and the Moon on its nonspherical shape.

- **Rigid Body**
- **A rigid body is a body in which the distance between any two of its constituent particles is constant under the action of many forces. In other words, a rigid body has fixed size and shape.**
- **The Kinetic energy, K of a rigid body rotating about a fixed point O is the sum of the individual kinetic energies of all the constituent particles.**
- **That is** $K = K_1 + K_2 + K_3 + \dots K_N$
- $$K = {}^1/_2\, m_1 v_1^2 + {}^1/_2\, m_2 v_2^2 + {}^1/_2\, m_3 v_3^2 + \dots {}^1/_2\, m_N v_N^2$$
-
- $$= \sum_{i=1}^{N} {}^1/_2\, m_i v_i^2$$
- **Recall that:** $v = r\omega$
- **Hence,**
- $$K = {}^1/_2\, \omega^2 \sum_{i=1}^{N} m_i r_i^2$$ **($\omega_i = \omega$ for all i) …constant**
- **Where the quantity** $\sum_{i=1}^{N} m_i r_i^2$ **is called the moment of inertia, I of the body**
- $$\therefore \quad \mathrm{K} = {}^1/_2\, I\, \omega^2$$
- **Comparing with** $K = {}^1/_2\, mv^2$ **in translational motion,**

- **the moment of inertial (I), plays a role analogous to that of mass *m*. However, while m is constant for a particular body, I depends on the location and direction of the axis of rotation as well as on the way the mass of the body is distributed**

The rotational kinetic energy of a rigid body is:

Rotational kinetic energy of a rigid body rotating around an axis

$$K = \frac{1}{2} I \omega^2$$

I: Moment of inertia of body for given rotation axis

ω: Angular speed of body

The moment of inertia, I, is obtained by multiplying the mass of each particle by the square of its distance from the axis of rotation and adding these products:

Moment of inertia of a body for a given rotation axis

$$I = m_1 r_1^2 + m_2 r_2^2 + \cdots = \sum_i m_i r_i^2$$

m_i: Masses of the particles that make up the body

r_i: Perpendicular distances of the particles from rotation axis

The SI unit of I is the kilogram-meter^2 (kg · m^2).

- **Calculation of Moment of Inertial of Bodies**
- **A rigid body is composed of a very large number of closely parked particles, so that the sum in the equation:**
- $\mathbf{I} = \sum_{i=1}^{N} m_i r_i^2$
- **maybe replaced by an integral.**
- **That is:**
- $\mathbf{I} = \sum_{i=1}^{N} m_i r_i^2 = \int r^2 \, dm$
- **If ρ is the density of the body, then:**
- $dm = \rho dv$ **and** $I = \int r^2 \rho dv = \rho \int r^2 \, dv$
- $\therefore \quad I = \rho \int r^2 \, dv$**, for a homogeneous body**

Moments of inertia of some common bodies

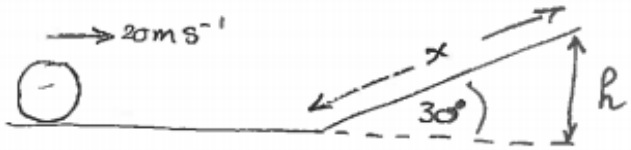
- **Radius of Gyration**
- **For all bodies rotating through any axis, the moment of inertial, I can be written as**
- $\mathbf{I} = mk^2$
- **Where k is the radius of gyration of the body about an axis.**
- **For example: For the above sphere rotating through its center,**
- $\therefore \quad k^2 = {}^2/_5 R^2$
- **Angular Momentum**
- **Angular momentum is defined as the first moment of momentum.**
- **For a particle with constant mass m, velocity v, momentum p = mv, and position vector $\vec{r}$ relative to the origin O of the inertial frame, we define angular momentum (L), as :**
- $$\vec{L} = \vec{r} \times \vec{p} = r \times mv$$
- **i.e** $L = \sum m\omega r^2$
- **(since** $I = \sum mr^2$**)**
- $L = I\omega$ **(Analogous to linear momentum P = mv)**
- **Hence:** $L = I\,{}^{d\theta}/_{dt}$

- **Torque on a Rigid Body**
- **The rotating effect on a body about an axis is called its Torque. Also, Torque is defined as the moment of force about an axis.**
- **i.e** $\vec{\tau} = \vec{r} \times \vec{F}$
- **Newton second law can then be written as:**
- $$\frac{d\vec{L}}{dt} = \vec{\tau}$$
- **i.e** $$\vec{\tau} = I\,\frac{d^2\theta}{dt^2} = I\frac{d\omega}{dt} = I\alpha$$
- **Torque of a force is the rate of change of angular momentum of a particle**
- **Kinetic Energy of a Rolling Object**
- **A rolling object of radius r has a moment of inertial** $I_g + \mathrm{m}r^2$, **using the parallel axis theorem.**
- $$K.E = \frac{1}{2}\,\mathrm{I}\omega^2 = \frac{1}{2}\left(I_g + \mathrm{m}r^2\right)\omega^2$$
- $$= \frac{1}{2}I_g\omega^2 + \frac{1}{2}\mathrm{m}r^2\omega^2$$
- $$= \frac{1}{2}I_g\omega^2 + \frac{1}{2}\mathrm{m}v^2$$
- **Hence , the K.E of a rolling object is equal to the sum of the K.Es of rotational and translational.**

- **Equilibrium of a Rigid Body**
- **The study of a rigid body in a state of equilibrium under the action of many forces is referred to as statics.**
- **Two conditions are required:**
- **The sum of all forces must be zero (for translational equilibrium)**
- $$\sum_i \boldsymbol{F}_i = \mathbf{0}$$
- **The sum of all torques in relation to any axis must be zero (For rotational equilibrium)**
- $$\sum_i \boldsymbol{\tau}_i = \mathbf{0}$$

- **Example:**
- **A 1.0 kg (solid) sphere rolling on a horizontal surface at 20 m s^{-1} comes to the base of an inclined plane which makes an angle of 30° with the horizontal**
- **Calculate the total kinetic energy of the ball when it is at the base of the incline.**

For a solid sphere $I = \frac{2}{5}MR^2$



$E_k\ (COG) = \frac{1}{2}mv^2$

$E_k\ (rot) = \frac{1}{2}I\omega^2 = \frac{1}{2}\cdot\frac{2}{5}mr^2\cdot\frac{v^2}{r^2} = \frac{1}{5}mv^2$

$\therefore E_k\ (tot) = \left(\frac{1}{2}+\frac{1}{5}\right)mv^2 = \frac{7}{10}mv^2 = \frac{7}{10}\times 1\times 20^2 = 280\ J$

Gravitation
Newton's Law of Gravitation

- Sir Isaac Newton, published his **law of gravitation** in 1687, which states that:
- **Every particle of matter in the universe attracts every other particle with a force that is directly proportional to the product of the masses of the particles and inversely proportional to the square of the distance between them.**

$$F_g = \frac{Gm_1m_2}{r^2}$$

where $G = 6.6742(10) \times 10^{-11}\ \mathrm{N{\cdot}m^2/kg^2}$.

- The force between two objects is equal and opposite (Newton's 3rd Law)
- Gravitational forces combine vectorially. If two masses exert forces on a third, the *total* force on the third mass is the vector sum of the individual forces of the first two. This is the principle of *superposition*.
- Gravity is always attractive.

WEIGHT

When calculating the gravitation pull of an object near the surface of the earth, the quantity GM_{earth}/R_{earth}^2 keeps appearing. It is more convenient to calculate this quantity and give it a name.

- $$F_g = m\left(\frac{GM_{\text{earth}}}{R_{\text{earth}}^2}\right) = m\,g$$

  i

  s

GRAVITATIONAL POTENTIALL ENERGY

Recall that our previous expression for gravitational potential energy near the surface of the earth is U = mgh. What is the gravitational potential energy as we move further away from the surface of the earth?

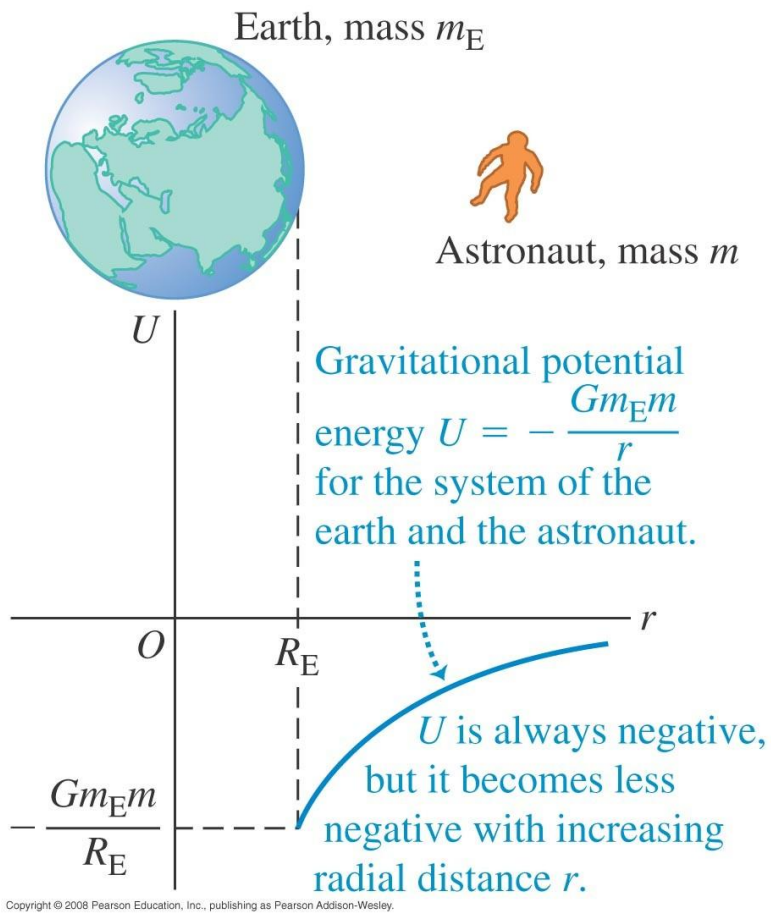
$$W_{\text{grav}} = \int_{r_1}^{r_2} F_r \, dr$$

$$F_r = -\frac{G m_E m}{r^2}$$

$$W_{\text{grav}} = G m_E m \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$$

$$U_{\text{grav}} = -\frac{G m_E m}{r}$$

Don’t confuse the expressions for gravitational force with



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ESCAPE VELOCITY

Use conservation of energy to find the escape velocity:

Esurface = E∞

Ksurface + Usurface = K∞ + U∞

$$mv^2 = -\frac{GMEmRE}{} = 0 - \frac{GMEm\infty}{}$$

$$= \sqrt{\frac{2GM_E}{R_E}}$$

By using Newton's 2nd Law and Newton's Law of Universal Gravitation, we can calculate the following:

$$v = \sqrt{\frac{G\,M_E}{r}}$$

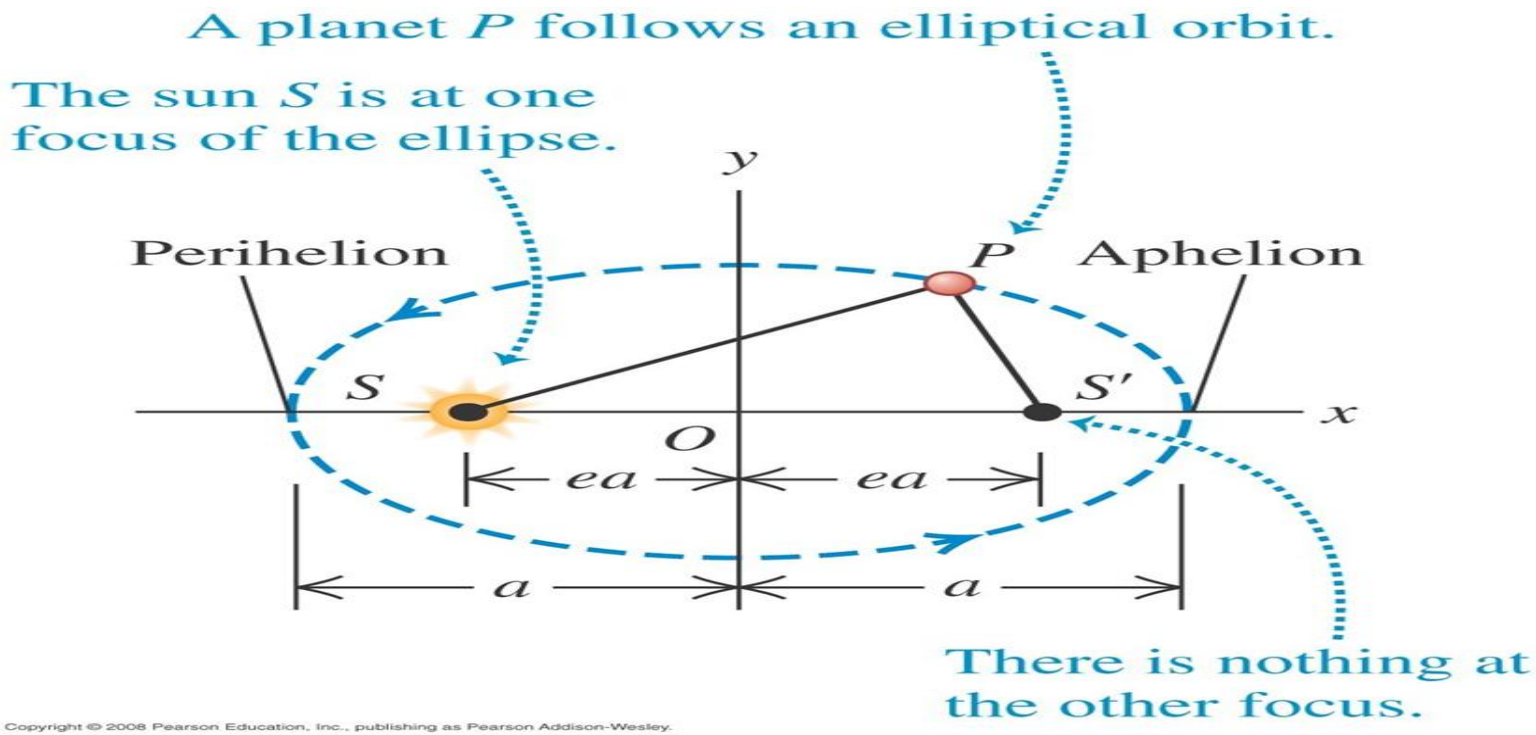
$$T = \frac{2\pi r}{v} = \frac{2\pi r^{3/2}}{\sqrt{G\,M_E}}$$

KEPLER'S LAW AND THE MOTION OF PLANETS

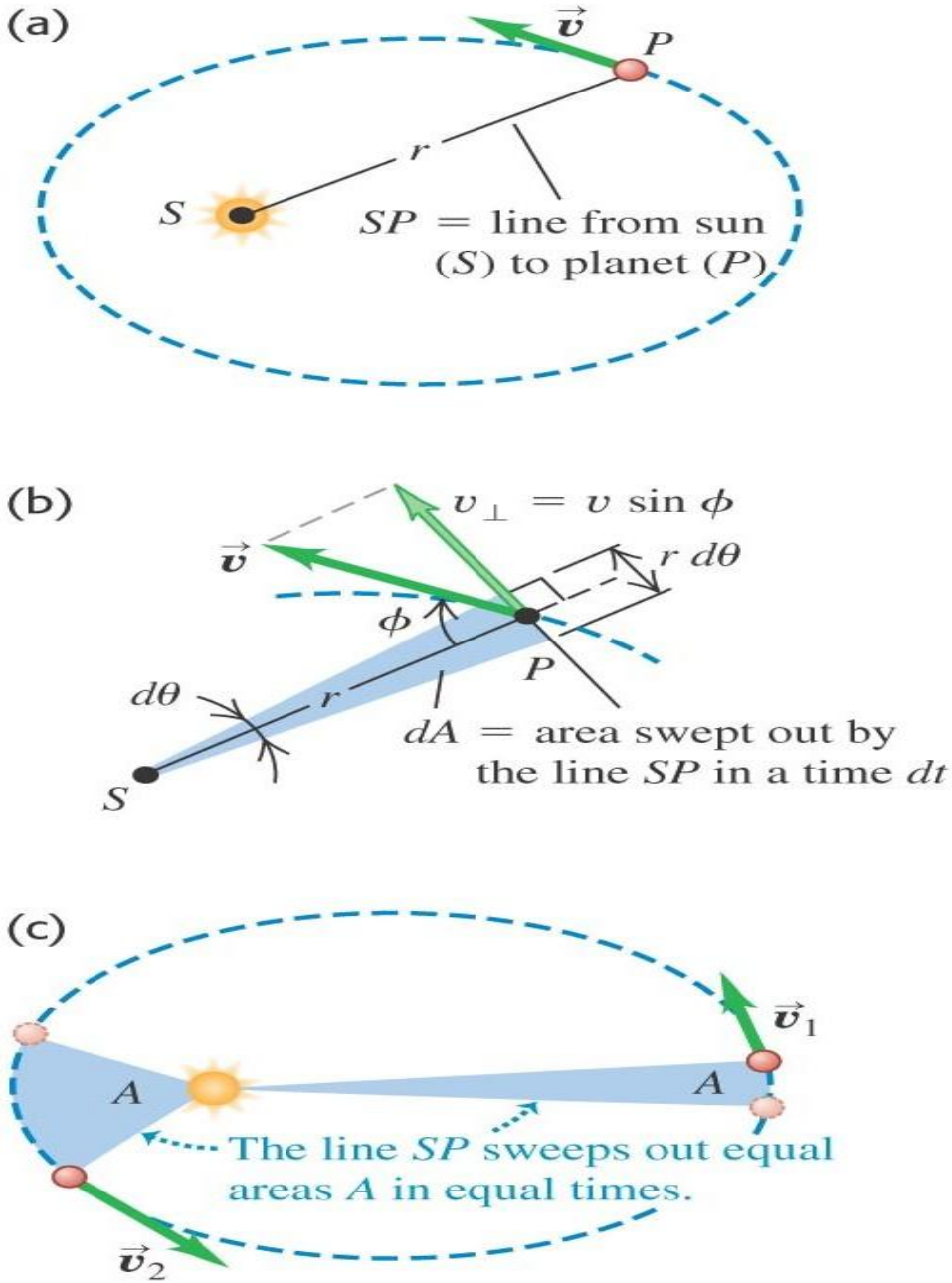
1st Law Each planet moves in an elliptical orbit, with the sun at one focus of the ellipse.

2nd Law A line from the sun to a given planet sweeps out equal areas in equal times.

3rd Law the periods of the planets are proportional to the powers of the major axis lengths of their orbits



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(a)
$\vec{v}$
P
r
S
SP = line from sun (S) to planet (P)
(b)
$v_\perp = v \sin \phi$
$\vec{v}$
$r\,d\theta$
ϕ
P
$d\theta$
r
S
dA = area swept out by the line SP in a time dt
(c)
$\vec{v}_1$
A
A
The line SP sweeps out equal areas A in equal times.
$\vec{v}_2$